higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 4

1 MARCH 2019

This marking guideline consists of 7 pages.

Copyright reserved

NOTE TO MARKERS

This marking guideline serves as a guideline and is NOT absolute. Consider alternative correct answers.

QUESTION 1: CALCULATIONS FOR CIVIL AND STRUCTURAL STEEL WORK

1.1 UDL = 3 000(1,5) = 4 500 (load is un- factored and needs to be multiplied by 1.5)
Reinforcement:
$N = 0,45\sigma_c bh + 0,73\sigma_y A_{sc}$
$4\ 500 \times 10^3 = 0,45(25)(500)(500) + 0,73(350)A_{sc}$
$4\ 500\ 000 = 2\ 812\ 500 + 255,5A_{sc}$
$4\ 500\ 000 - 2\ 812\ 500 = 255,5A_{sc}$
$1\ 687\ 500 = 255,5A_{sc}$
$1\ 687\ 500/255,5 = A_{sc}$
$A_{top} = A_{bottom} = 6\ 605\ mm^2/2 = 3\ 303\ mm^2$ (7)

1.2 $T_r = \phi A f_y$
$T_r = 0,9(2,21 \times 10^3)(450)$
$T_r = 895\ kN$ (3)

1.3 $C_r = \phi A f_y(1+ \lambda^{2,68})^{-1/1,34}$
$= 0,9(9,29 \times 10^3)(350)(1 + 1,6^{2,68})^{-1/1,34}$
$= 1\ 523\ kN$ (3)

1.4 λ:

$$\lambda = \frac{KL}{r}\sqrt{\frac{fy}{\pi^2 E}}$$

$$\lambda = \frac{2(6\ 000)}{64,6}\sqrt{\frac{350 \times 10^6}{\pi^2(200 \times 10^9)}}$$

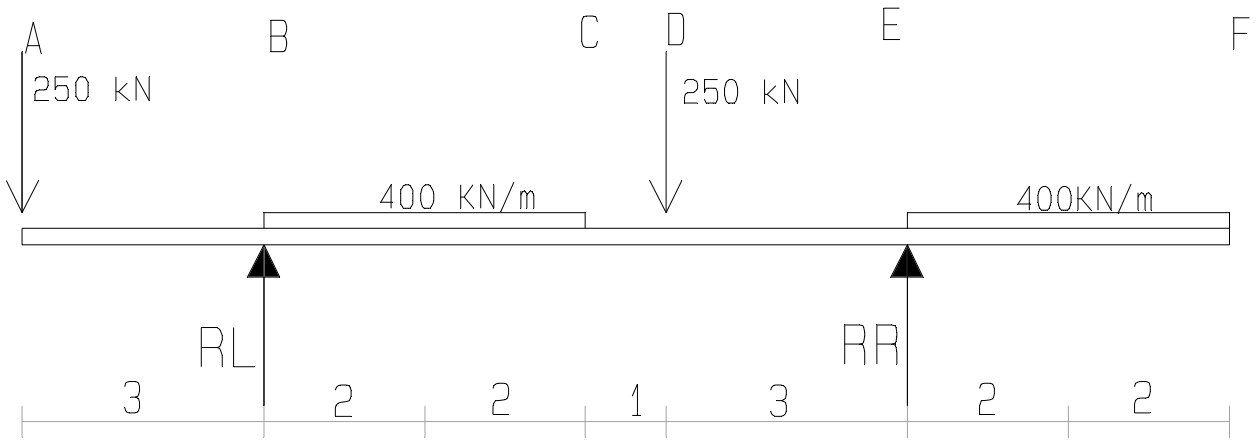
$$= 2,47\ \sqrt{}$$

Compressive resistance:

$C_r = \phi A f_y(1+ \lambda^{2,68})^{-1/1,34}$
$C_r = 0,9(9,29 \times 10^3)(350)(1+ 78,2^{2,68})^{-1/1,34}$
$= 0,48\ kN$ (5)

Copyright reserved

1.5



1.5.1 Take moments about RL:

ACW = CW
(RR × 8) + (250 × 3)✓= (400 × 4 x 2) + (250 × 5) + (400 ×4 x 10)

R = 2462,5 kN✓

Take moments about RR:

CW = ACW
(L × 8) + (400 x 4 × 2) = (250 × 3) + (400 x 4 × 6) + (250 × 11)
L = 1237,5 kN

Test: upwards F = downwards F
2462,5 + 1237,5 = 1600 + 1600 + 250 +250✓

3700 = 3700✓ (8)

1.5.2 At A: -250 down = -250 kN✓
At B: -250 + 1237,5 = 987,5 kN✓

At C: +987,5 – 1600 = -612,5 kN✓
At D: -612,5 - 250 = -862,5 kN✓
At E: -862,5 + 2462,5 = 1600 kN ✓
At F: +1600 – (4 × 400) = 0✓ (6)
[32]

QUESTION 2: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

2.1
- Slump cone or slump mould
- Solid waterproof base
- Tamping rod (steel) (3)

2.2
- Roller which resists vertical forces
- Hinge support which resists horizontal and vertical forces (2 × 2) (4)

2.3
- Ignore the self-weight and calculate the maximum bending moment that the beam has to resist.
- Use both design equations to determine the value of Z_e and $Z_{pl.}$
- Use the value of Z_e and Z_{pl} to select an appropriate section.
- Calculate the self-weight of the section to determine the new maximum bending moment (UDL).
- Use the properties of this section and the equations in the table showing H- and I-section properties to determine the class of the section.
- Select the appropriate design equation.
- Determine whether the beam will be able to resist the bending moment by determining the maximum resisting moment that the beam can handle. (7)

2.4 2.4.1
(a) mild steel
(b) high-yield steel
(2 × 1) (2)

2.4.2
(a) Top
(b) Bottom
(c) Each way
(d) Alternate bars reversed
(e) Staggered
(f) Alternately
(g) Top of grating
(7 × 1) (7)

2.5
- A pile cap is a thick concrete mat that rests on concrete or timber piles driven into soft or unstable ground to provide a suitable stable foundation.
- It usually forms part of the foundation of a building, typically a multi-storey building, structure or support base for heavy equipment. (2)

[25]

QUESTION 3

3.1 G
3.2 F
3.3 B
3.4 A
3.5 H
3.6 E
3.7 D
3.8 C

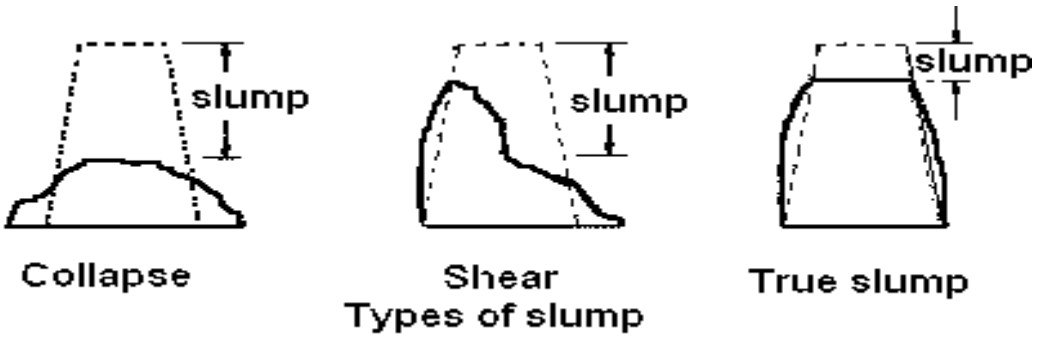
(8 × 1) **[8]**

QUESTION 4: PROPERTIES OF STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

4.1 A slump test is performed to check the consistency of freshly made concrete. The slump test is done to make sure a concrete mix is workable.
A cube test measures the compressible cube strength of the concrete and relates directly to the required design strength specified by the designer.

(4)

4.2



Types of slump

In a shear slump the top portion of the concrete shears off and slips sideways.✓✓ On the other hand, if one half of the cone slides down an inclined plane, the slump is said to be a shear slump.✓✓ In a collapsed slump, the concrete collapses completely. A collapsed slump will generally mean that the mix is too wet or that it is a high workability mix, for which a slump test is not appropriate.✓✓

(6)
[10]

TOTAL: 75

Copyright reserved